

Light-cone-quantized QCD in 1+1 dimensions, revisited

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I. INTRODUCTION

In 1990, Hornbostel, Brodsky and Pauli diagonalized the light-cone Hamiltonian of $SU(N)$ gauge theory in one space and one time dimension [1]. They computed the hadronic spectrum and wavefunctions across couplings, colors, and baryon number at resolutions $2K = 16\text{--}24$, in minutes on the mainframes of the day. Convergence was controlled by a single extrapolation formula. The paper became the standard reference for discretized light-cone quantization (DLCQ), and its figures are still the picture most physicists carry of QCD in two dimensions at finite N .

The method has a long lineage. DLCQ was introduced by Pauli and Brodsky [2, 3] and first applied to the Schwinger model [4]; reviews are Refs. [5, 6]. The same framework later reached finite temperature in QED_{1+1} , including thermodynamics and the analogue of the R ratio [7, 8]. Light-cone wavefunctions feed a much wider class of observables than spectra.

Nor did the program stop in two dimensions. Positronium and quarkonium served as 3+1 DLCQ test cases early [9], front-form QED resolved the positronium spin multiplets [10, 11], and parallel light-front solvers later delivered precision true-muonium spectra [12, 13]. Higher-dimensional QCD advanced through transverse-lattice methods [14], supersymmetric DLCQ and its relatives [15], and basis light-front quantization [16], which now produces quarkonium spectra in full 3+1 dimensions [17]. Two-dimensional adjoint QCD remains under active light-front study with basis-function methods [18]. Ref. [19] maps the community program. Each branch of this tree validates, at some point in its development, against the two-dimensional calculation revisited here.

The audience for that calculation has changed. Today the model is the standard benchmark for Hamiltonian lattice methods, tensor networks, and quantum simulation, and light-front quantization is itself being developed as a basis for quantum computation [20–22]. Claims of quantum advantage on 1+1d gauge theories are structurally weak, because classical methods handle these theories so well [23]. This paper sharpens that point with a ledger. A direct qubit encoding of the Fock space diagonalized here occupies $2N_c$ fermionic modes per momentum: roughly 70 logical qubits to re-run the 1990 calculation, and 300–800 for the resolutions and color groups in this paper. The classical calculation completes in seconds to minutes on

one workstation.

The value of the old code to quantum simulation is therefore not as a target but as a reference standard. It supplies controlled, error-budgeted continuum numbers against which equal-time hardware results can be scored. A reference is only as good as its validation, and DLCQ and the equal-time formulations hardware uses are inequivalent quantizations. Confirming the limits in which they agree is part of this paper’s job. The one regime where they currently disagree is mapped in Sec. X.

The model also has open questions of its own. Light-front and equal-time methods measure different chiral-limit behavior at finite N (Sec. X). Large- N analyses predict a tetraquark bound state near the two-meson threshold, and the prediction awaits a finite- N test [24, 25].

This paper re-does the 1990 calculation with thirty-six years of hardware and of error analysis. The error analysis matters more. We reproduce every figure and the table of that paper, at the original parameters and again at resolutions up to $2K \simeq 100$. Two independent implementations of the same Hamiltonian run against each other: the preserved Fortran of the original program, and a validated open-source port. The original controlled its errors with one extrapolation series and a parenthesis convention; we carry a four-component error budget with held-out validation (Sec. VII). Its weak-coupling entries were limited by the momentum grid in a way no resolution can repair. We identify the mechanism, quantify it without free parameters, and reach the chiral region instead with van de Sande’s improved Hamiltonian [26] (Sec. VIII).

Two sections have no 1990 counterpart. Section IX searches for states dominated by four- and six-parton Fock sectors: tetraquark and hexaquark candidates. Section X compares against the modern record: integrability at large N , tensor-network and quantum-hardware spectra at $N = 2, 3$, and precision Schwinger-model benchmarks.

We keep the structure of Ref. [1] deliberately, section for section where possible, so each figure can be laid beside its 1990 counterpart. Sections II and III fix notation and recall the Hamiltonian. Section IV describes what changed computationally. Sections V–VI reproduce and extend the 1990 results. The remainder is new.

II. LIGHT-CONE QUANTIZATION

We follow the conventions of Ref. [1] throughout. Light-cone coordinates are $x^\pm = x^0 \pm x^1$, with x^+ the time. The mass-shell condition $p^+p^- = m^2$ makes p^- the energy

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of a quantum carrying longitudinal momentum $p^+ > 0$. Positivity of p^+ is the structural advantage of the scheme. No combination of quanta with positive momenta can add up to the vacuum's $p^+ = 0$, so the Fock vacuum is an exact eigenstate of the full Hamiltonian. Every basis state is built from quanta that belong to the hadron, not to vacuum fluctuations.

Discretization places the theory in a box of length $2L$ in x^- with antiperiodic fermions, so momenta come in half-odd-integer units $k_n^+ = 2\pi n/L$. The dimensionless resolution

$$K = \frac{L}{2\pi} P^+ \quad (1)$$

is a positive integer, and a basis state is a color-singlet partition of K among quanta. (In code units we track $2K$, whose parity matches the parton number.) At $K = 3$ the available momentum fractions are the partitions (3), (2, 1) and (1, 1, 1). An equal-time discretization at the same volume admits unbounded sets of positive and negative momenta.

The resolution is also the infrared regulator: the smallest momentum fraction is $x = k/K \sim 1/K$. The continuum limit $L \rightarrow \infty$ at fixed P^+ is the single limit $K \rightarrow \infty$. When the $1/K$ endpoint cutoff dominates the error is the subject of Secs. VII and VIII. Everything else — mode expansions, constrained fields, the construction of P^- — is unchanged from Ref. [1], and we do not repeat it.

III. SU(N) IN 1+1 DIMENSIONS AND THE IMPROVED HAMILTONIAN

The theory is SU(N) gauge theory coupled to one fundamental fermion of mass m ,

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} + \bar{\psi} (i\not{D} - m) \psi, \quad (2)$$

in one space and one time dimension. The coupling g carries dimensions of mass, so m/g is the only physical parameter. In the gauge $A^+ = 0$ the gluon carries no dynamical degrees of freedom: a constraint fixes A^- , and integrating it out leaves quarks interacting through an instantaneous linear potential in x^- . Confinement is kinematic in two dimensions, in QED₁₊₁ as much as here. Half of the fermion components are also constrained. The physical content is one spinless quark field per color.

The light-cone Hamiltonian takes the form

$$P^- = m^2 \sum_{n,c} \frac{a_{nc}^\dagger a_{nc}}{k_n^+} + g^2 H_I. \quad (3)$$

H_I contains the normal-ordered four-quark vertices of the instantaneous potential, with color structure $\delta_{c_1 c_3} \delta_{c_2 c_4} - \frac{1}{N} \delta_{c_1 c_2} \delta_{c_3 c_4}$ and a $1/(\Delta k^+)^2$ kernel. It also contains a one-body “self-induced inertia”: the normal-ordering remnant of the exchange, diagonal in the Fock basis. Explicit matrix elements are in Ref. [1]. Our implementations of them are public (Appendix D).

III.1. The endpoint problem and the improved Hamiltonian

Near the endpoints $x \rightarrow 0, 1$ the continuum bound-state wavefunction behaves as $\phi \sim x^a$, with the exponent set by the transcendental condition [1, 27]

$$1 - \pi a \cot(\pi a) = \frac{\pi m^2}{g^2 C_F}, \quad C_F = \frac{N^2 - 1}{2N}. \quad (4)$$

As $m/g \rightarrow 0$ the exponent $a \rightarrow 0$, and the endpoint region $x \lesssim e^{-1/a}$ shrinks below any fixed grid. A uniform grid with smallest fraction $1/K$ then replaces the endpoint integral $\int dx x^{2a-1} \sim 1/2a$ by a logarithm $\ln K$. No reachable resolution recovers it (Sec. VII). This is a property of the grid, not of light-cone quantization: a continuum ’t Hooft solver on a uniform mesh, sharing no code with the discretized theory, fails the same way.

Improvement is the standard cure for discretization diseases in lattice QCD: improved nonrelativistic actions [28] and highly improved staggered quarks [29] both remove the leading cutoff artifacts at the level of the action rather than the fit. Van de Sande’s improved Hamiltonian [26] plays the same role here and treats the disease at its source. The self-induced inertia and the Coulomb exchange carry separate, canceling endpoint singularities. The improvement replaces the free-field self-energy by one computed against the interacting endpoint behavior of Eq. (4), so the residual kernel vanishes when either momentum approaches an endpoint.

Within an existing DLCQ code the improvement costs almost nothing, for two reasons. First, it is exactly diagonal: swapping the self-energy table $\sigma_{\text{std}} \rightarrow \sigma_{\text{imp}}$ leaves every interaction kernel, and the standard path, untouched bit for bit. Second, the color weight it needs is $\langle -T_a \cdot T_c \rangle$ at zero momentum transfer, which the color-contraction machinery already computes. The identity $\sum_{c \neq a} \langle -T_a \cdot T_c \rangle = C_F$ holds exactly in our implementation at every N and parton number we use. We enforce it as a gate rather than assume it.

One rule attaches to everything that follows. The two Hamiltonians approach the continuum along different paths: the improved spectrum extrapolates in a plain $1/K$ series [26], while the standard spectrum requires the fractional-power ladder of Ref. [1], Eq. (27) there. No fit in this paper crosses the two bases. Sec. VII quantifies what happens when one does.

IV. THE NUMERICAL METHOD

The 1990 program worked in three steps [1]. It generated all color-singlet partitions of K , removed overcomplete directions by diagonalizing the inner-product matrix, and diagonalized the Hamiltonian in the surviving basis. Free and interacting parts were stored separately, so a change of coupling is a rescaling rather than a rebuild. Nothing in that architecture needed changing. What changed is what it runs on.

We run two implementations against each other. The first is the original Fortran, kept unmodified as a historical reference. The second is an open-source Python port whose matrix builds are verified bit-identical to their own independent reference path (Appendix D). The two solvers agree at the level of the arithmetic, not the plot. One defect of the original era was found along the way: both codebases silently capped the color-contraction tables, which corrupted matrix elements above certain basis sizes with no warning. Both paths now raise.

Three observations take the reachable resolution from $2K = 16\text{--}24$ to $2K \simeq 100$ on a laptop-class machine. First, the inner-product matrix is exactly block diagonal in parton configuration, verified by brute-force color enumeration. Orthonormalization therefore never needs the dense n^2 object that was the binding memory constraint. Second, the Hamiltonian in the weeded basis is sparse and gets sparser with K , so an iterative eigensolver pays off once the norm is blocked. Third, a blockwise basis projects the free part to an exact diagonal, which removes the last dense intermediate. None of this approximates the physics: the sparse path must reproduce the dense reference to solver precision, and the standard-Hamiltonian path remains bit-identical to what the code has always produced.

Resolution is bounded above on purpose. At $2K = 100$ the solver reproduces its recorded reference values in seconds. Past it, the color path returns spurious negative eigenvalues below an otherwise intact spectrum; the released code raises instead of returning them. Wrong-parity requests also raise. Mesons need even $2K$ and baryons need $2K \equiv N \bmod 2$; the failure mode this closes is an empty spectrum returned in a tenth of a second, which reads as a fast successful run.

The 1990 paper noted its largest runs took “one to three minutes of CPU on an IBM 3081.” The corresponding statement here: the full table of extrapolated masses, some eight hundred solves, rebuilds from nothing in about six minutes on one idle workstation. The binding constraint is no longer time or memory. It is the endpoint physics of Sec. VII.

V. WAVE FUNCTIONS AND STRUCTURE FUNCTIONS

VI. MASSLESS MESONS AND BARYONS IN THE $m/g \rightarrow 0$ LIMIT

At $m = 0$ the spectrum contains towers of exactly massless hadrons. The construction is Sec. VI of Ref. [1] and survives unchanged, so we record only what this paper uses. The color currents V_k^a obey a Kac–Moody algebra, and the interacting part of P^- is a quadratic form in them. States created purely by currents, and for baryon number by a composite baryon operator, are annihilated by it. Massless mesons, baryons, and multi-hadron states therefore exist at every resolution K , not merely in the

limit. The massless baryon’s valence distribution is closed-form,

$$q(x) = N(N-1)(1-x)^{N-2}, \quad (5)$$

and collapses at $N = 2$ to the flat meson distribution. For $U(N)$ the extra $U(1)$ current promotes these states to free bosons at $M^2 = g^2/2\pi$: the Schwinger mechanism, and the sharpest statement of how $SU(N)$ and $U(N)$ differ in two dimensions.

Three later results lean on this section. The identically-zero last row of Table ?? is exact and K -independent; our solvers reproduce it at every resolution as a structural check. The chiral fits of Sec. VIII use masslessness as the fixed endpoint any $M^2(m/g)$ form must respect. And the exotics search of Sec. IX inherits a boundary condition: multi-hadron states are exactly massless at $m = 0$, so every binding energy we measure must vanish in the chiral limit. A candidate whose Δ fails to do so is diagnosing its own systematics, not new physics.

VII. CONVERGENCE AND THE ERROR BUDGET

VIII. THE CHIRAL REGION

IX. TETRAQUARK AND HEXAQUARK CANDIDATES

X. COMPARISON WITH MODERN METHODS

The 1990 paper compared against the two external results then available: ’t Hooft’s large- N solution and Hamer’s finite-lattice $SU(2)$ spectrum [30]. The modern record is far richer. This section is organized by what each method can check.

X.1. Large N and integrability

The ’t Hooft equation is now solved to arbitrary precision: it is equivalent to a TQ-Baxter system [31, 32], which makes the large- N meson spectrum exact for our purposes. We rescale our finite- N masses to ’t Hooft’s normalization (Appendix C) and find the expected $1/N^2$ approach, updating Fig. 8(b) of Ref. [1] with error bars. The same comparison shows what large N misses at strong coupling: light baryons, and the $U(1)$ distinction of Sec. VI. The original argued both points; here the residuals are quantified.

X.2. Lattice

Hamer’s finite-lattice masses remain the only equal-time spectrum across the full coupling range at $N = 2$. We

compare against his published Table 1. Its stated weak-coupling uncertainties are of order the values themselves; digitizing his figure would suggest a precision his own table disclaims, so we do not. Within those bars, agreement is complete. That includes the $N = 2$ identity $M_{\text{bar}} = M_{\text{mes}}$, which his data was the first to exhibit. Modern overlap-fermion work confirms the expected infrared behavior at $(N_c, N_f) = (2, 1)$ and $(3, 1)$ [33], though without spectrum tables to compare digit for digit.

X.3. Tensor networks and quantum hardware

Matrix-product-state calculations now reach SU(2) [34, 35] and SU(3) [36, 37] in one dimension. Variational quantum hardware has produced SU(2) meson and baryon masses [38], with SU(3) preparations following [39]. These are staggered-fermion computations at finite lattice spacing, so the honest comparison is between matched continuum extrapolations, not raw numbers. Where the published works quote continuum values, we tabulate them beside ours. Precision Schwinger-model benchmarks [40, 41] calibrate what tensor networks achieve at U(1); our own equal-time exact-diagonalization cross-checks reproduce them in our conventions (Appendix D). No public code exists for any of the SU(N) tensor-network calculations above. The pipeline released with this paper is, to our knowledge, the first reproducible finite- N spectrum code in either quantization.

X.4. The chiral-exponent tension

One discrepancy in this landscape is structural. We state it as the open question it is. Every light-front determination of the finite- N chiral exponent finds $M^2 \propto (m/g)^\alpha$ with $\alpha \rightarrow 1$: this work, lightcone conformal truncation [42], and $1/N$ -corrected 't Hooft perturbation theory [43]. The one equal-time measurement, an MPS study of SU(2) over $m/g \in [0.1, 0.4]$ [35], reports $\nu = 0.700(29)(11)$ for $M \propto (m/g)^\nu$. That is $\alpha \simeq 1.4$: consistent with bosonization, and several standard deviations from the light-front value.

The methods differ in exactly the assumption all light-front schemes share. The light-front vacuum is trivial, and the vacuum is what an anomalous chiral exponent is about. How equal-time vacuum structure re-emerges as the light cone is approached was analyzed by Hornbostel himself [44]; the question this tension raises is that analysis made quantitative. The equal-time side is not yet decisive either. Its measured window spans a factor of four in m/g , not a decade. And no SU(N) analogue of the additive staggered mass shift [41] has been published or applied, though the shift is the same order in the lattice spacing as the mass being fitted. Resolving the tension requires an equal-time computation in the same chiral window with that renormalization under control. It is beyond the published record, and beyond this paper.

XI. PROSPECTS

Five extensions are within reach of this machinery, in rough order of readiness.

QCD coupled to QED with physical charges. The color-contraction tables already compute both Fierz structures, and the U(1) exchange reuses the $-1/N$ index structure with different weights. $H = H_{\text{QCD}} + (e^2/g^2) H_{\text{QED}}$ is therefore a reweighting of matrix elements already built, and one build serves every charge ratio. The validation ladder is clean: bit-identity at $e = 0$, the exact Schwinger mass at the U(1) point, isospin restoration. One obstruction is physical. Gauss' law on a circle admits only neutral states without a zero-mode subtraction, so the neutron and π^0 come free and the proton does not.

The U(N) switch. The same reweighting in the pure-U(1) limit exposes the SU(N)/U(N) split of Sec. VI numerically. The thesis [45] plots the SU(2)–U(2) comparison the 1990 article never showed. Reproducing it would close the last unreproduced published figure of this program.

Zero modes. Our antiperiodic quantization removes the $k^+ = 0$ mode by fiat. Where zero modes have been included dynamically in two-dimensional models, low-lying masses shift by up to $\sim 20\%$ [46]. Their absence is a genuine systematic of unknown size for our absolute masses. It is not the cause of the weak-coupling failure: a continuum solver with no zero-mode truncation reproduces that failure exactly (Sec. VII). Collective-variable treatments that tensor a zero-mode sector onto the Fock basis exist and fit our solver architecture.

Hadronic tensors and finite temperature. Light-cone wavefunctions are raw material for more than spectra. Structure functions are already in hand (Sec. V), and DLCQ has historically reached finite-temperature QED₁₊₁, including the analogue of the R ratio [7, 8]. Two-current correlators and hadronic tensors are the quantities recent quantum simulations target at the hundred-qubit scale [47]. They are natural DLCQ observables, computable here with controlled errors at a classical cost of seconds. A DLCQ pipeline for $W^{\mu\nu}$ would hand the quantum-simulation program the reference standard argued for in Sec. I.

An equal-time computation in the same chiral window. The sharpest open discrepancy this paper documents will not be settled by more K : light-front methods find $\alpha \rightarrow 1$ where the one equal-time measurement suggests $\alpha \approx 1.4$ (Sec. X). Settling it needs an equal-time calculation with a controlled additive mass renormalization at finite N , over a decade in m/g rather than a factor of four. The missing ingredient is the SU(N) analogue of the staggered-fermion mass shift known for the Schwinger model [41]. Deriving it is, in our view, the single most valuable piece of new work available in this corner of field theory, and the natural subject of a companion study.

Appendix A: Error-budget definitions

Every extrapolated mass carries an uncertainty built from four components, combined in quadrature. All four are measured. None is fit.

form: The spread of the extrapolated mass over the order of the fit series: for the standard Hamiltonian, two through five terms of the Eq. (27) ladder of Ref. [1]. It is the dominant fit component, and it is irreducible. The extrapolate does not converge in the order, and the two defensible order-selection criteria prefer opposite ends of the admissible range (Appendix B). The budget therefore spans the range rather than choosing.

window: The spread over contiguous sub-windows of the resolution range at fixed order. It is correlated with *form*; the ratio of the two runs near one half across the weak-coupling entries. The quadrature sum still reproduces the joint order-times-window ensemble spread to a few percent at $m/g \leq 0.2$, so the budget does not double count there. At strong coupling it mildly over-covers, which we accept.

trunc: The change in the extrapolate when the Fock-space truncation is raised by one $q\bar{q}$ pair. This component grows with K at weak coupling. It is measured on the analysis window actually used, never quoted from a smaller probe.

endpoint: The captured-fraction systematic of the grid. With endpoint exponent a from Eq. (4), a grid of resolution K captures a fraction $1 - K^{-2a}$ of the endpoint mass integral. The one-sided upward systematic on $M(0)$ is

$$\delta_{\text{ep}} = M(0) \left[\frac{1}{1 - K^{-2a}} - 1 \right], \quad (\text{A1})$$

with no free parameters. It enters the total as an uncertainty, never as a correction: the model is too crude to shift central values, and too well verified to ignore. The same expression accounts for ninety-five percent of the measured standard-versus-improved gap across a factor of thirty in coupling (Sec. VIII). At $m/g \leq 0.2$ it dominates every other component. It is what makes the weak-coupling bars of Table ?? honest.

The solver's own floor, at relative 10^{-13} from the eigen-solver tolerance, is carried through and never visible at quoted precision. The improved Hamiltonian runs the same machinery with the plain- $1/K$ series in place of the Eq. (27) ladder. Its budget has no endpoint component; removing that error is what the improvement is (Sec. III.1).

Two validation habits back the budget rather than extend it. Held-out extrapolation fits the low-resolution points and predicts the highest two, a basis-independent

check of the fit form. And each quoted mass prints the fraction supplied by extrapolation rather than data, so a reader can see where the fit interpolates and where it asserts.

Appendix B: Approaches measured not to work

Every strategy below was implemented and measured against the standard-Hamiltonian weak-coupling problem of Sec. VII, then rejected. We record them because each is the obvious next idea. Each also fails invisibly from inside the fit, which is the property that makes the underlying problem dangerous. Numbers refer to the Table ?? channels at $m/g \leq 0.2$ unless stated.

More resolution.: The remainder falls only as K^{-a} with $a \rightarrow 0$. Halving the uncertainty bracket at $m/g = 0.1$ requires raising K by a factor of order $10^{4.9}$; at our smallest couplings the factor exceeds any conceivable computation. Resolution buys nothing below the validity boundary. That is the point.

A confluent fit basis.: Replacing the near-degenerate $\{K^{-1}, K^{-(1+a)}\}$ pair by a confluent basis spans the same space. The extrapolated mass is identical to one part in 10^{14} .

Committing to a single series order.: Fixing the number of terms hides the order dependence rather than removing it: the next admissible term still moves the extrapolate by several percent. Held-out resolution and consistency with the known chiral behavior prefer opposite ends of the admissible range, which is what an undetermined model order looks like. The budget spans the ladder instead. Constrained fits [48] are the principled alternative and would need priors on coefficients the endpoint physics leaves unconstrained; we did not pursue them.

A validity floor on the fit windows.: Raising the smallest admitted resolution shrinks the quoted spread severalfold while walking the central value monotonically, by many times the shrunken bar. Variance is traded for bias, silently.

Fitting the endpoint exponent freely.: $M_0 - C K^{-p}$ is unidentifiable when the true p is small. K^{-p} is nearly constant, M_0 and C degenerate, and the fit lands an order of magnitude off with a residual of 10^{-7} . A perfect fit to the wrong answer.

Summing the tail analytically.: A Hurwitz- ζ resummation of the increment series validates at strong coupling. It diverges where it is needed.

Adding an explicit K^{-a} column.: With a fixed from Eq. (4), the extrapolated masses land several times above the published values, and the fitted chiral

exponent stays pinned to the grid artifact. Inside a joint $(m/g, K)$ fit the same column produces variant spreads of hundreds of percent and negative masses.

The common failure mode deserves the emphasis: every rejected strategy improved some internal diagnostic while degrading the answer. Residuals, held-out predictions, and spread statistics all reward flexibility inside the reachable window, and the reachable window does not contain the physics (Sec. VII). Where we quote weak-coupling numbers from the standard Hamiltonian, the bar carries the endpoint systematic in full. Where we need the chiral region, we change the Hamiltonian, not the fit.

Appendix C: Units and conventions

Masses are quoted as M/g unless stated otherwise, and m/g is the quark mass in units of the coupling. Our natural combination is $g^2 N/2\pi$; 't Hooft's classic solution uses $g^2 N/\pi$. The large- N comparison in Sec. X therefore plots $(2\pi/N)^{1/2} M/g$ against $(2\pi/N)^{1/2} m/g$, as Fig. 8(b) of Ref. [1] did. Spectra shown against the coupling use $\lambda = [1 + \pi m^2/g^2]^{-1/2}$, which sends the free limit to $\lambda \rightarrow 0$ and the chiral limit to $\lambda = 1$, and the dimensionless mass $M^2/(m^2 + g^2/\pi)$. Both axes match the 1990 figures. In code units the resolution is the integer $2K$, whose parity matches the parton number. Single-parton momenta are odd integers, so a structure function lives on $x = k/2K$ with measure $dx = 1/K$.

Two conventions of Ref. [1] are traps. Both bit us, so we state them plainly. First, the numbers printed in its Table I are $M^2/(m^2 + g^2/\pi)$, although the column headers read M/g . The two agree at no coupling. We established the identification by recomputing both quantities at the 1990 parameters and matching digit patterns; our

transcription carries the correction with full provenance. Every comparison in this paper converts to genuine M/g , and the build recomputes the comparison in the table's native M^2 units as a cross-check (Appendix D).

Second, the parenthesized digits in that table are not error bars. They record the magnitude of the last term retained in the extrapolation series, a convergence diagnostic the authors themselves trusted only for $m/g \gtrsim 0.2$. Where this paper quotes an uncertainty, it is the full budget of Sec. VII. Where it quotes the 1990 numbers, the parenthesis keeps its original meaning.

Captions in this paper state N , B , $2K$ or the fit window, m/g , the Fock-space truncation, and the Hamiltonian for every panel. Where the original figure left a parameter unstated, we cite the recovered value and the evidence for it inline.

Appendix D: Data and code availability

Both solvers are public: the preserved 1988-era Fortran, unmodified, and the Python port. The repository also carries every analysis script, the digitized markers of all 1990 figure panels with their axis-calibration provenance, and the per-figure data files. Each data file records in its header the exact command that regenerates it.

Every number in this paper's prose is a macro generated from those data files by a single extractor. The extractor records the repository revision and input hashes alongside the values. Every figure is built by a single driver that reads cached solver output and refuses to solve; a missing input surfaces as an explicit run command, not a silent recomputation. The equal-time exact diagonalization used for the Schwinger-model cross-checks of Sec. X is included.

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- [1] K. Hornbostel, S. J. Brodsky, and H. C. Pauli, Light Cone Quantized QCD in (1+1)-Dimensions, *Phys. Rev. D* **41**, 3814 (1990).
 - [2] H. C. Pauli and S. J. Brodsky, Solving Field Theory in One Space One Time Dimension, *Phys. Rev. D* **32**, 1993 (1985).
 - [3] H. C. Pauli and S. J. Brodsky, Discretized Light Cone Quantization: Solution to a Field Theory in One Space One Time Dimensions, *Phys. Rev. D* **32**, 2001 (1985).
 - [4] T. Eller, H. C. Pauli, and S. J. Brodsky, Discretized Light Cone Quantization: The Massless and the Massive Schwinger Model, *Phys. Rev. D* **35**, 1493 (1987).
 - [5] S. J. Brodsky, G. McCartor, H. C. Pauli, and S. S. Pinsky, The Challenge of light cone quantization of gauge field theory, *Part. World* **3**, 109 (1993).
 - [6] S. J. Brodsky, H.-C. Pauli, and S. S. Pinsky, Quantum chromodynamics and other field theories on the light cone, *Phys. Rept.* **301**, 299 (1998), [arXiv:hep-ph/9705477](#).
 - [7] S. Strauss and M. Beyer, Light front QED(1+1) at finite temperature, *Phys. Rev. Lett.* **101**, 100402 (2008), [arXiv:0805.3147 \[hep-th\]](#).
 - [8] S. Strauss and M. Beyer, Massive light-front Schwinger model at finite temperature, *Prog. Part. Nucl. Phys.* **62**, 535 (2009).
 - [9] M. Krautgartner, H. C. Pauli, and F. Wolz, Positronium and heavy quarkonia as testing case for discretized light cone quantization. 1., *Phys. Rev. D* **45**, 3755 (1992).
 - [10] U. Trittman, Front form QED in (3+1)-dimensions: The Spin multiplet structure of the positronium spectrum at strong coupling, in *New Nonperturbative Methods and Quantization of the Light Cone* (1997) pp. 89–96, [arXiv:hep-th/9706055](#).
 - [11] U. Trittman, On the role of the annihilation channel in front form positronium, (1997), [arXiv:hep-th/9705072](#).
 - [12] H. Lamm and R. F. Lebed, True muonium ($\mu^+\mu^-$) on the light front, *J. Phys. G* **41**, 125003 (2014), [arXiv:1311.3245 \[hep-ph\]](#).
 - [13] H. Lamm and R. F. Lebed, Nonperturbative True Muonium on the Light Front with TMSWIFT, *Few Body Syst.* **57**, 663 (2016), [arXiv:1601.05771 \[hep-ph\]](#).

- [14] M. Burkardt and S. Dalley, The Relativistic bound state problem in QCD: Transverse lattice methods, *Prog. Part. Nucl. Phys.* **48**, 317 (2002), [arXiv:hep-ph/0112007](#).
- [15] J. R. Hiller, Nonperturbative light-front Hamiltonian methods, *Prog. Part. Nucl. Phys.* **90**, 75 (2016), [arXiv:1606.08348 \[hep-ph\]](#).
- [16] J. P. Vary, H. Honkanen, J. Li, P. Maris, S. J. Brodsky, A. Harindranath, G. F. de Teramond, P. Sternberg, E. G. Ng, and C. Yang, Hamiltonian light-front field theory in a basis function approach, *Phys. Rev. C* **81**, 035205 (2010), [arXiv:0905.1411 \[nucl-th\]](#).
- [17] Y. Li, P. Maris, X. Zhao, and J. P. Vary, Heavy Quarkonium in a Holographic Basis, *Phys. Lett. B* **758**, 118 (2016), [arXiv:1509.07212 \[hep-ph\]](#).
- [18] U. Trittman, Solving two-dimensional adjoint QCD with a basis-function approach, *Phys. Rev. D* **109**, 016004 (2024), [arXiv:2307.15212 \[hep-th\]](#).
- [19] B. L. G. Bakker *et al.*, Light-Front Quantum Chromodynamics: A framework for the analysis of hadron physics, *Nucl. Phys. B Proc. Suppl.* **251-252**, 165 (2014), [arXiv:1309.6333 \[hep-ph\]](#).
- [20] M. Kreshchuk, W. M. Kirby, G. Goldstein, H. Beauchemin, and P. J. Love, Quantum simulation of quantum field theory in the light-front formulation, *Phys. Rev. A* **105**, 032418 (2022), [arXiv:2002.04016 \[quant-ph\]](#).
- [21] M. Kreshchuk, S. Jia, W. M. Kirby, G. Goldstein, J. P. Vary, and P. J. Love, Simulating Hadronic Physics on NISQ devices using Basis Light-Front Quantization, *Phys. Rev. A* **103**, 062601 (2021), [arXiv:2011.13443 \[quant-ph\]](#).
- [22] T. Li, X. Guo, W. K. Lai, X. Liu, E. Wang, H. Xing, D.-B. Zhang, and S.-L. Zhu (QuNu), Partonic collinear structure by quantum computing, *Phys. Rev. D* **105**, L111502 (2022), [arXiv:2106.03865 \[hep-ph\]](#).
- [23] H. Lamm, No Quantum Utility from Hadron Masses? No, Quantum Utility from Hadron Masses!, (2026), [arXiv:2603.00946 \[hep-lat\]](#).
- [24] H. Sazdjian, Meson scattering and tetraquarks in two-dimensional QCD, *Phys. Rev. D* **112**, 054029 (2025), [arXiv:2504.19809 \[hep-ph\]](#).
- [25] W. Lucha, D. Melikhov, and H. Sazdjian, Tetraquark and two-meson states at large N_c , *Eur. Phys. J. C* **77**, 866 (2017), [arXiv:1710.08316 \[hep-ph\]](#).
- [26] B. van de Sande, Convergence of discretized light cone quantization in the small mass limit, *Phys. Rev. D* **54**, 6347 (1996), [arXiv:hep-ph/9605409](#).
- [27] G. 't Hooft, A Two-Dimensional Model for Mesons, *Nucl. Phys. B* **75**, 461 (1974).
- [28] G. P. Lepage, L. Magnea, C. Nakhleh, U. Magnea, and K. Hornbostel, Improved nonrelativistic QCD for heavy quark physics, *Phys. Rev. D* **46**, 4052 (1992), [arXiv:hep-lat/9205007](#).
- [29] E. Follana, Q. Mason, C. Davies, K. Hornbostel, G. P. Lepage, J. Shigemitsu, H. Trottier, and K. Wong (HPQCD, UKQCD), Highly improved staggered quarks on the lattice, with applications to charm physics, *Phys. Rev. D* **75**, 054502 (2007), [arXiv:hep-lat/0610092](#).
- [30] C. J. Hamer, SU(2) Yang-Mills Theory in (1+1)-dimensions: A Finite Lattice Approach, *Nucl. Phys. B* **195**, 503 (1982).
- [31] F. Ambrosino and S. Komatsu, 2d QCD and integrability. Part I. 't Hooft model, *JHEP* **02**, 126, [arXiv:2312.15598 \[hep-th\]](#).
- [32] A. Litvinov and P. Meshcheriakov, Meson mass spectrum in QCD2 't Hooft's model, *Nucl. Phys. B* **1010**, 116766 (2025), [arXiv:2409.11324 \[hep-th\]](#).
- [33] N. Karthik, R. Narayanan, and S. A. Narayanan, Approaching the conformal WZW behavior in the infrared limit of two-dimensional massless QCD: A lattice study, *Phys. Rev. D* **109**, 054502 (2024), [arXiv:2312.13929 \[hep-lat\]](#).
- [34] P. Silvi, E. Rico, M. Dalmonte, F. Tschirsich, and S. Montangero, Finite-density phase diagram of a $(1+1) - d$ non-abelian lattice gauge theory with tensor networks, *Quantum* **1**, 9 (2017), [arXiv:1606.05510 \[quant-ph\]](#).
- [35] M. C. Bañuls, K. Cichy, J. I. Cirac, K. Jansen, and S. Kühn, Efficient basis formulation for 1+1 dimensional SU(2) lattice gauge theory: Spectral calculations with matrix product states, *Phys. Rev. X* **7**, 041046 (2017), [arXiv:1707.06434 \[hep-lat\]](#).
- [36] P. Silvi, Y. Sauer, F. Tschirsich, and S. Montangero, Tensor network simulation of an SU(3) lattice gauge theory in 1D, *Phys. Rev. D* **100**, 074512 (2019), [arXiv:1901.04403 \[quant-ph\]](#).
- [37] T. Hayata, Y. Hidaka, and K. Nishimura, Dense QCD₂ with matrix product states, *JHEP* **07**, 106, [arXiv:2311.11643 \[hep-lat\]](#).
- [38] Y. Y. Atas, J. Zhang, R. Lewis, A. Jahanpour, J. F. Haase, and C. A. Muschik, SU(2) hadrons on a quantum computer via a variational approach, *Nature Commun.* **12**, 6499 (2021), [arXiv:2102.08920 \[quant-ph\]](#).
- [39] R. C. Farrell, I. A. Chernyshev, S. J. M. Powell, N. A. Zemlevskiy, M. Illa, and M. J. Savage, Preparations for quantum simulations of quantum chromodynamics in 1+1 dimensions. I. Axial gauge, *Phys. Rev. D* **107**, 054512 (2023), [arXiv:2207.01731 \[quant-ph\]](#).
- [40] M. C. Bañuls, K. Cichy, K. Jansen, and J. I. Cirac, The mass spectrum of the Schwinger model with Matrix Product States, *JHEP* **11**, 158, [arXiv:1305.3765 \[hep-lat\]](#).
- [41] R. Dempsey, I. R. Klebanov, S. S. Pufu, and B. Zan, Discrete chiral symmetry and mass shift in the lattice Hamiltonian approach to the Schwinger model, *Phys. Rev. Res.* **4**, 043133 (2022), [arXiv:2206.05308 \[hep-th\]](#).
- [42] N. Anand, A. L. Fitzpatrick, E. Katz, and Y. Xin, Chiral limit of 2d QCD revisited with lightcone conformal truncation, *JHEP* **01**, 189, [arXiv:2111.00021 \[hep-th\]](#).
- [43] I. V. Kochergin, 1/N corrections in QCD₂: small mass limit and threshold states, *JHEP* **02**, 073, [arXiv:2405.04031 \[hep-th\]](#).
- [44] K. Hornbostel, Nontrivial vacua from equal time to the light cone, *Phys. Rev. D* **45**, 3781 (1992).
- [45] K. Hornbostel, *The Application of Light-Cone Quantization to Quantum Chromodynamics in One-Plus-One Dimensions*, Ph.D. thesis, Stanford University (1988).
- [46] A. S. Mueller, A. C. Kalloniatis, and H.-C. Pauli, Effect of zero modes on the bound state spectrum in light cone quantization, *Phys. Lett. B* **435**, 189 (1998), [arXiv:hep-th/9803204](#).
- [47] E. J. Gustafson, I. B. Hernandez, F. Herren, E. M. Jeeninga, M. Kharchenko, H. Lamm, E. J. Steeno, R. S. Van de Water, and M. L. Wagman, Hadronic tensors in $\mathbb{Z}_2$ lattice gauge theory on 101 qubits (2026), fermilab report FERMILAB-PUB-26-0459-STUDENT-T, in preparation.
- [48] G. P. Lepage, B. Clark, C. T. H. Davies, K. Hornbostel, P. B. Mackenzie, C. Morningstar, and H. Trottier (HPQCD), Constrained curve fitting, *Nucl. Phys. B Proc. Suppl.* **106**, 12 (2002), [arXiv:hep-lat/0110175](#).